# Software Requirements Specification (SRS)

## KodeX – Competitive Programming Practice Platform

Ver 1.0

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# 1. Introduction

## 1.1 Purpose

This document outlines the specifications for KodeX, a competitive programming practice platform enabling users to solve coding problems with automated server-side execution and detailed verdict analysis.

## 1.2 Scope

- Practice coding problems by difficulty and tags
- Secure online code execution using Judge0 API
- Automated verdict system (AC/WA/TLE/MLE/CE)
- Submission tracking and analytics
- Leaderboards and ranking system
- Problem setter and admin dashboards
- Responsive web application

## 1.3 Intended Audience

- Stakeholders
- Developers
- Testers
- Designers
- System Administrators

## 1.4 Product Features

- User registration & login (JWT-based authentication)
- Role-based access control
- Problem creation with visible and hidden test cases
- Problem filtering and search
- Monaco-based online IDE
- Multi-language support
- Dual execution engine
- Per-test-case verdict breakdown
- Submission history & analytics
- Leaderboard and badge system
- Discussion forum with moderation
- Plagiarism detection
- Notification system
- Dark/Light mode

## 1.5 Architecture Overview

- Frontend: React.js, Tailwind CSS, Monaco Editor
- Backend: Node.js, Express.js
- Database: MongoDB Atlas
- Code Execution: Judge0 CE API
- Hosting: AWS / Render

## 1.6 Future Enhancements

- AI-based coding hints
- Code complexity analyzer
- Mobile application
- Gamification seasons
- Multi-language UI support

## 1.7 User Stories for KodeX Platform

| ID | User Role | Wants To | So That |
|---|---|---|---|
| US001 | User | Register using email | Access platform securely |
| US002 | User | Login to account | Solve problems |
| US003 | User | Submit code | Receive automated verdict |
| US004 | User | View submission history | Track improvement |
| US005 | User | View leaderboard | Compare ranking |
| US006 | Problem Setter | Create problems | Add challenges |
| US007 | Admin | Manage users | Maintain system integrity |
| US008 | Moderator | Remove comments | Maintain discussion quality |
| US009 | User | Search and filter problems | Find relevant questions |
| US010 | User | Earn badges | Stay motivated |

## 1.8 Functional Requirements (FR)

| ID | Title | Description | Priority | Risk | Risk % |
|---|---|---|---|---|---|
| FR001 | User Registration | Allow user registration with validation | 1 | Medium | 20% |
| FR002 | Authentication | JWT based login system | 1 | High | 35% |
| FR003 | RBAC | Role based access control | 1 | High | 40% |
| FR004 | Problem Creation | Setters create problems with test cases | 1 | Medium | 30% |
| FR005 | Dual Execution | Execute user & expected solution | 1 | Critical | 65% |
| FR006 | Verdict System | Generate AC/WA/TLE/MLE/CE | 1 | High | 50% |
| FR007 | Submission History | Store submission attempts | 2 | Low | 15% |
| FR008 | Leaderboard | Rank users | 2 | Medium | 30% |
| FR009 | Plagiarism Detection | Detect similar code | 1 | Critical | 70% |
| FR010 | Discussion Forum | Threaded comments system | 2 | Medium | 25% |

## 1.9 Non-Functional Requirements (NFR)

| ID | Title | Description |
|---|---|---|
| NFR001 | Performance | Response time < 2 seconds |
| NFR002 | Concurrency | Support 500 concurrent users |
| NFR003 | Secure Execution | Sandboxed execution |
| NFR004 | Data Security | Password hashing & HTTPS |
| NFR005 | Availability | 99% uptime |
| NFR006 | Scalability | Cloud horizontal scaling |

## 1.10 KodeX Platform - Project Plan

**Overview**

- **Project Duration:** 10–12 Weeks
- **Development Methodology:** Agile (Scrum-based with 1-week sprints)

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**Phase 1: Requirement Analysis & Planning (Week 1)**

**Deliverables:**

- Finalized Software Requirements Specification (SRS)
- Prioritized Feature List (MVP vs Advanced Features)
- Technology Stack Confirmation (MERN + Judge0)
- Project Roadmap and Gantt Chart
- Risk Assessment Document

**Tasks:**

- Stakeholder meetings and requirement gathering
- Finalize user stories for work distribution
- Define MVP scope
- Identify functional & non-functional requirements
- Create initial risk analysis
- Setup Jira, GitHub, Figma, Slack
- Define branching strategy and coding standards

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**Phase 2: Design (Week 2)**

**Deliverables:**

- UI Wireframes and High-Fidelity Prototypes
- Database Schema Design (MongoDB Collections)
- ER Diagram

- API Specification Document
- System Architecture Diagram

**Tasks:**

- Design user flows (registration → problem solving → submission)
- Create problem page layout
- Design leaderboard & dashboard UI
- Define REST API endpoints
- Review designs with stakeholders
- Finalize database relationships

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**Phase 3: Backend Development (Week 3–4)**

**Deliverables:**

- REST APIs (Node.js + Express)
- JWT-based Authentication System
- Role-Based Access Control (RBAC)
- Problem Management APIs
- Submission & Verdict Engine
- Judge0 API Integration
- Plagiarism Detection Logic (basic version)

**Tasks:**

- Setup backend project structure
- Implement authentication & authorization
- Create problem CRUD APIs
- Implement test case validation logic
- Integrate Judge0 for code execution
- Develop output comparison module

- Implement submission storage
- Write unit tests (Jest)
- Setup logging & error handling

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**Phase 4: Frontend Development (Week 5–6)**

**Deliverables:**

- Responsive Web Application (React)
- Monaco Editor Integration
- Problem Repository Page
- Submission Result Interface
- Leaderboard Page
- User Dashboard

**Tasks:**

- Setup React project structure
- Implement authentication UI
- Connect frontend with backend APIs
- Build online IDE interface
- Implement problem filtering & search
- Create submission history page
- Design badge & analytics components
- Implement dark/light mode toggle

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**Phase 5: Testing & Bug Fixing (Week 7)**

**Deliverables:**

- Test Cases Documentation
- Manual Testing Report

- API Testing Report (Postman)
- Security Testing Summary
- Performance Testing Results

**Tasks:**

- Perform functional testing of all modules
- Test edge cases in code execution
- Validate verdict accuracy
- Perform load testing for concurrent users
- Fix integration bugs
- Optimize slow APIs
- Verify security vulnerabilities (JWT, injection risks)

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**Phase 6: Deployment (Week 8)**

**Deliverables:**

- Production Deployment
- Dockerized Backend Services
- SSL Configuration (HTTPS)
- Cloud Hosting Setup (AWS / Render)
- Database Backup Configuration
- Monitoring & Logging Setup

**Tasks:**

- Setup production server
- Configure environment variables
- Enable HTTPS & security headers
- Deploy frontend and backend
- Configure MongoDB Atlas backups

- Setup monitoring tools
- Perform final smoke testing

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**Phase 7: MVP Launch (Week 9)**

**Deliverables:**

- Live Production Platform
- Admin Dashboard Active
- Leaderboard & Analytics Functional
- Plagiarism Monitoring Enabled
- User Onboarding Guide

**Tasks:**

- Soft launch with limited users
- Monitor system performance
- Collect initial feedback
- Fix high-priority bugs
- Announce official launch
- Prepare maintenance plan

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**Phase 8: Stabilization & Enhancements (Week 10–12)**

*(Optional buffer phase to match 10–12 week duration)*

**Deliverables:**

- Performance Improvements
- Advanced Analytics Dashboard
- UI/UX Refinements
- Documentation Finalization

**Tasks:**

- Optimize database queries
- Improve caching mechanisms
- Refactor inefficient modules
- Improve accessibility
- Final documentation & demo preparation

**Tools & Technologies**

| Area | Tools / Stack |
|---|---|
| Frontend | React.js (Web), Tailwind CSS, Monaco Editor |
| Backend | Node.js, Express.js |
| Database | MongoDB Atlas |
| DevOps | Docker, AWS/Render, GitHub CI |
| Design | Figma |
| Project Management | Jira, Confluence |
| Communication | Slack, Google Meet |
| Testing | Postman, Jest, OWASP ZAP |

**Timeline Overview**

| Phase | Duration (Weeks) | Sprint(s) |
|---|---|---|
| Req. Analysis & Planning | 1 | Sprint 1 |
| Design | 1 | Sprint 2 |
| Backend Development | 2 | Sprint 3–4 |
| Frontend Development | 2 | Sprint 5–6 |
| Testing & Bug Fixing | 1 | Sprint 7 |
| DevOps & Deployment | 1 | Sprint 8 |
| MVP Launch | 1 | Sprint 9 |

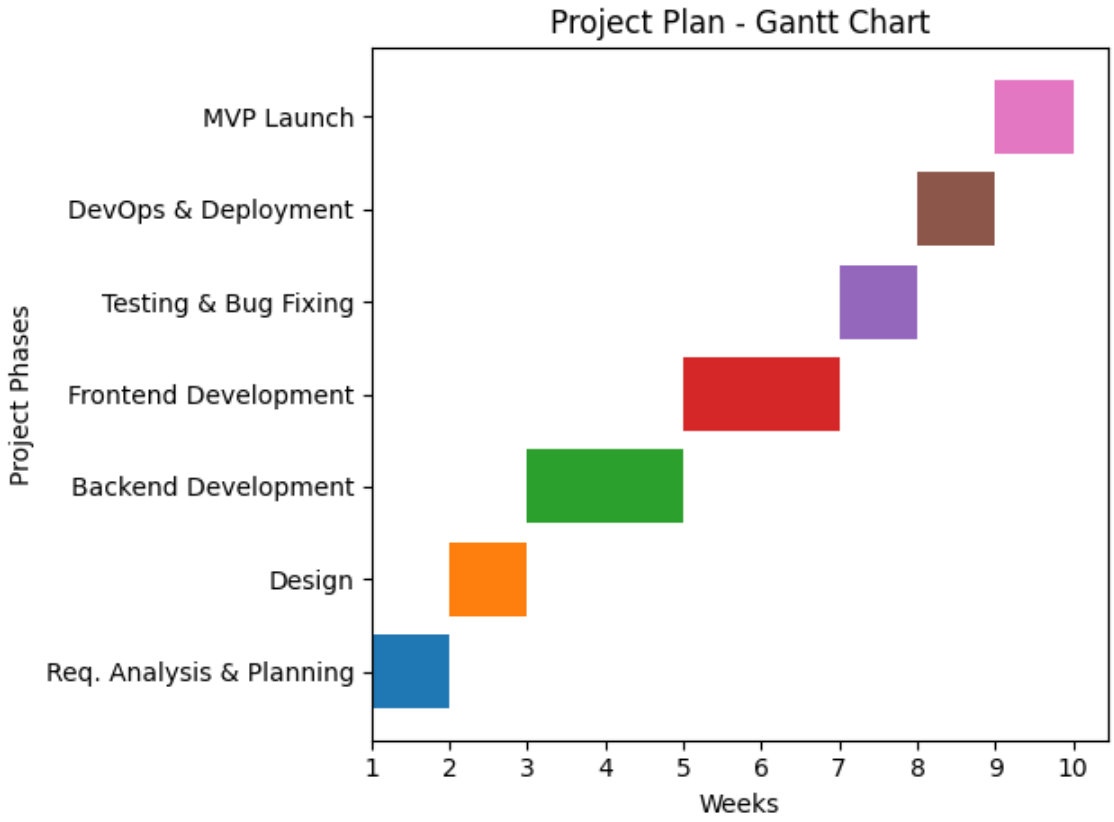


Figure 1: Project Plan - Gantt Chart